Who should we invite first?

Prioritising low-cost cognitive-health outreach for adults 45+

Background

Who we are

A public-health agency with a $250,000 budget for low-cost cognitive-health outreach (memory screening invitations, lifestyle counselling, follow-up referral)

Why this matters

Self-reported memory worsening - an early signal worth checking. As the population ages, demand for memory screening and support grows, but our outreach budget does not.



Aim of this exercise

- Prioritise older adults for outreach
- Identify modifiable factors associated with reported decline

Today's baseline

Random invitations reach people with decline at the base rate: About 21 in every 100.

Behavioural Risk Factor Surveillance System

356,158
rows, with 283 columns

73,448
respondents with a valid answer (2025 LLCP file)

20.6%
report cognitive decline (15,166 Yes)

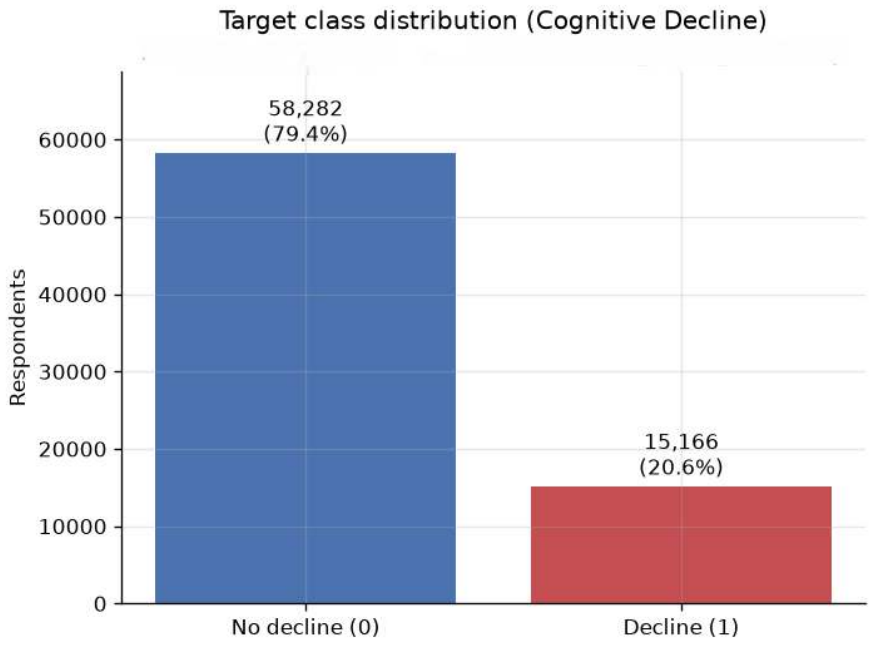
14
predictors

Target definition

"In the past 12 months, have thinking or memory difficulties happened more often or got worse?"
Yes = 1, No = 0; Don't know, refused and blank dropped;
118 under-45 records removed.

Known limitations

Optional module
Self-reported, not a dementia diagnosis
Cross-sectional: Associations only
Imbalanced classes (≈ 4:1)



Predictors

Non-modifiable risk markers

- Age
- Sex
- Prior heart attack
- Prior stroke

Manageable clinical conditions

- High blood pressure
- High cholesterol
- Diabetes
- Depression

Potentially modifiable health factors

- Poor mental-health days
- Sleep
- Physical activity
- Smoking
- Alcohol
- BMI

"Modifiable" means potentially changeable. The data is cross-sectional, so it does not show that changing these factors would prevent decline.

Binary classification

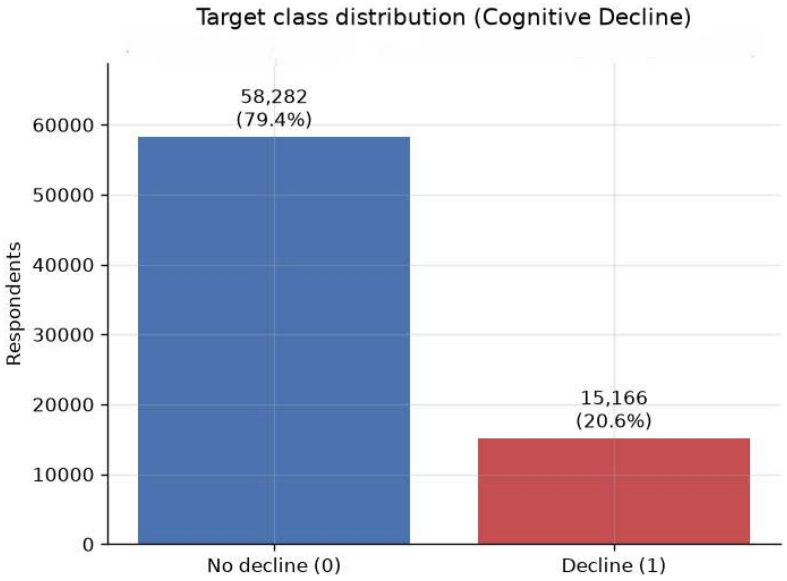
1 **Supervised**

Labelled target: CIMEMLO1, self-reported worsening memory in the past 12 months

2 **Binary Classification**

Outcome is Yes (1) vs No (0)

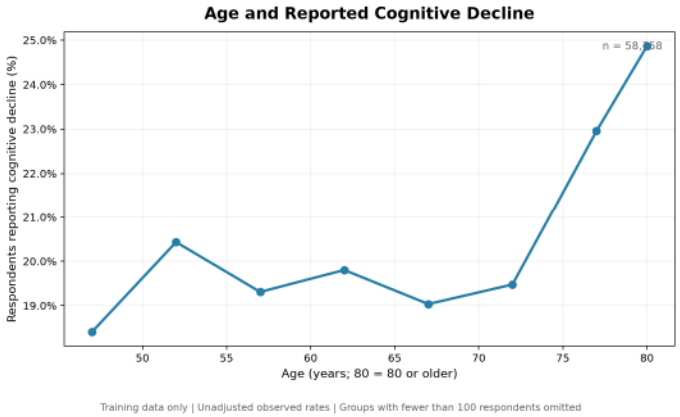
EDA: Class imbalance

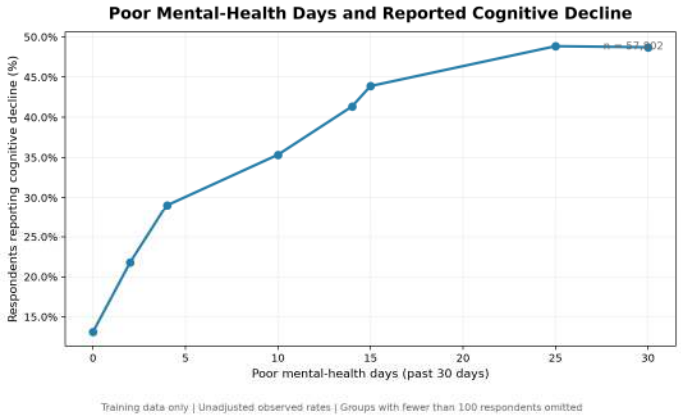


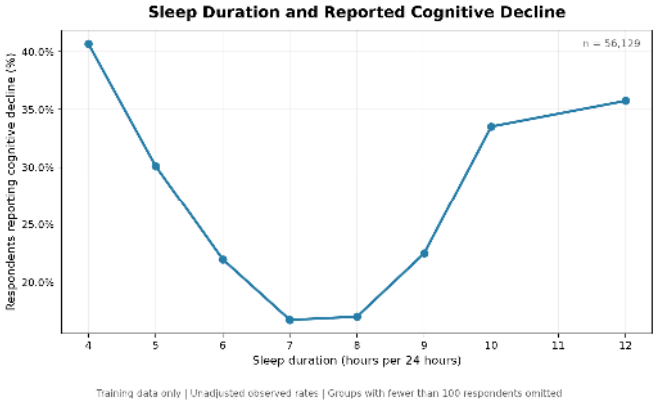
MODELLING RESPONSE

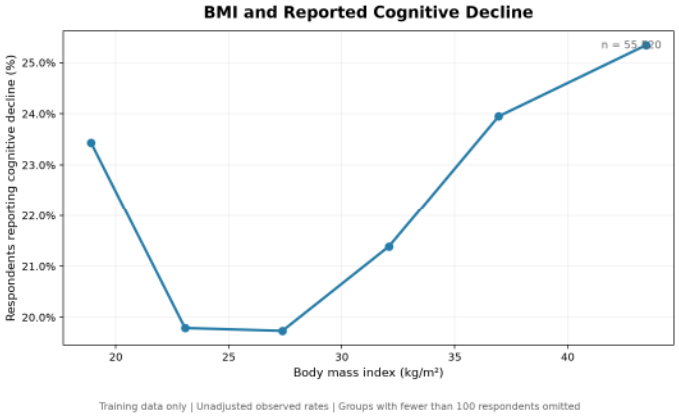
Stratified train/test split; use Average Precision and recall rather than accuracy alone

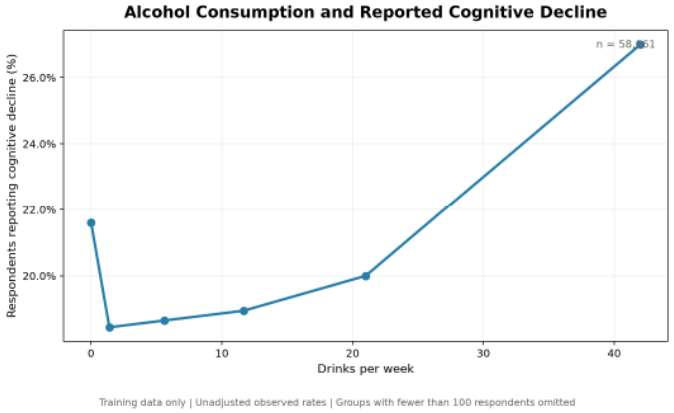
EDA: Non-linear data











Data handling

1

Clean (before split)

- Sentinels 7/9 recoded → NaN or 999 'Unknown'
- BMI ÷100
- Sleep capped at 16 h; mental-health 88 → 0
- Pregnancy-only BP/diabetes → 'No'
- Ages 45+ only

2

Split first

- Stratified 80/20 before anything is learned from the data
- Train 58,758 · Test 14,690

3

Missing values

- 'Was-missing' flag for each continuous variable
- Median imputation (robust to skew)
- Categorical gaps kept as 'Unknown' level

4

Spline terms

- Age, sleep, mental-health days, BMI, drinks
- Smooth curves instead of straight lines
- Used in logistic regression; tree models split natively

5

Scale & encode

- Standard scaling of continuous terms
- One-hot encoding of 10 categoricals

Models attempted

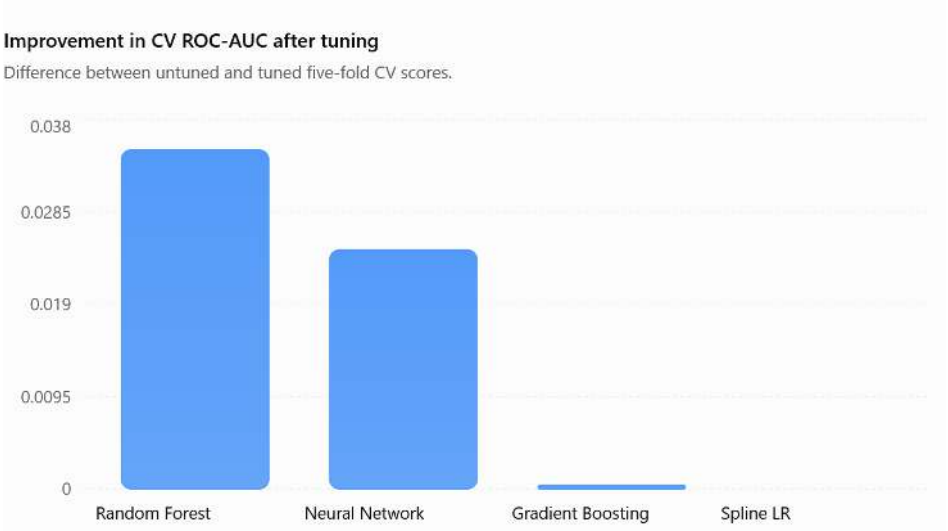
Every model outperforms the dummy's baseline

Model (5-fold CV, train only)	ROC-AUC	AP	Settings
Dummy (majority class)	0.500	0.206	Majority-class baseline
Spline Logistic Regression	0.731	0.431	C = 1, 4 spline knots, minority-class weighting
Random Forest	0.693	0.375	100 trees, unlimited depth, min leaf=1
Gradient Boosting	0.732	0.439	100 trees, depth 3, lr 0.1
Neural network (MLP) *	0.701	0.396	(100,), α=0.0001, lr=0.001

* The default neural network reached its maximum iteration limit without convergence

Hyperparameter Tuning and Findings

Model	Grid searched	Winner	Finding
Spline Logistic Regression	C: 0.001, 0.01, 0.1, 1, 10	C = 1	CV AUC 0.731
Random Forest	trees 200/300, depth 10/20, leaf 1/5	300, depth 10, leaf 5	CV AUC 0.728
Gradient Boosting	trees 100/200, depth 3/5, lr 0.05/0.1	200, depth 3, lr 0.05	CV AUC 0.732
Neural network (MLP)	architecture and alpha; separate four-rate sweep	(32,16), alpha 0.01, lr 0.01	CV AUC 0.726

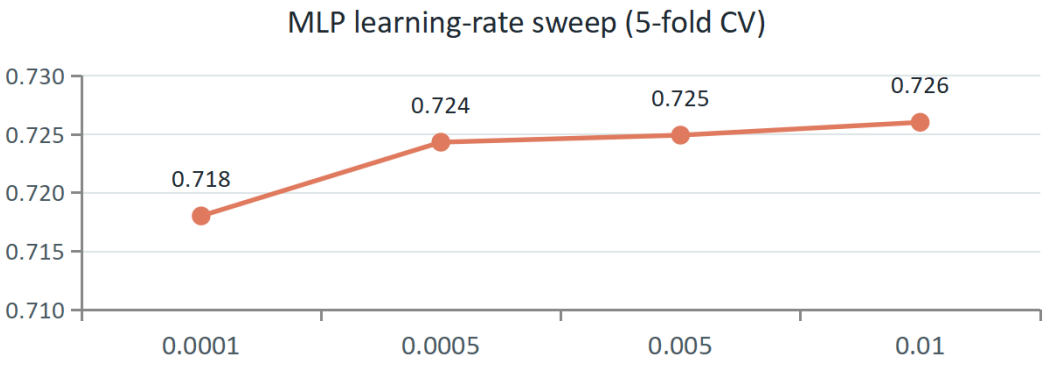


Clearest effects

Limiting tree depth and requiring at least five observations per leaf encourages broader, less specific decision rules.
This can improve generalisation to unseen observations.
Increasing the number of trees can also stabilise predictions.

Hyperparameter Tuning and Findings

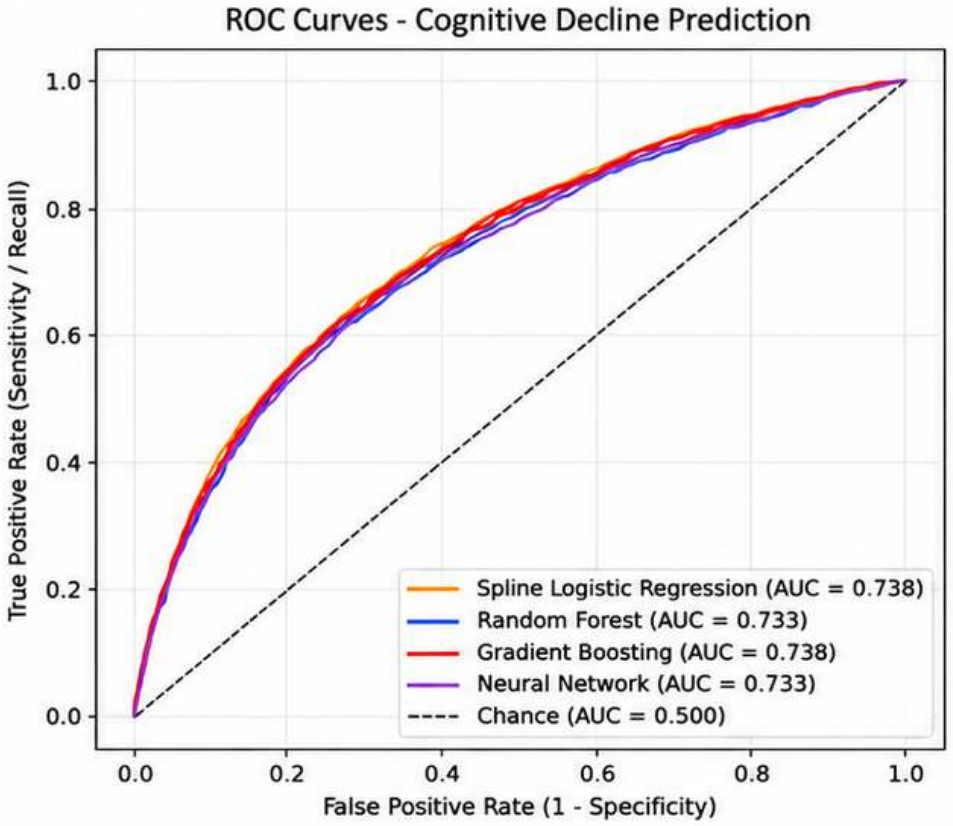
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Clearest effects

Feature representation mattered more than hyperparameters: splines gave +0.015 AUC to LR. Within grids, the MLP learning rate 0.0001 → 0.0005 gave +0.006; elsewhere tuning moved AUC ~0.01 at most.

Held-out test

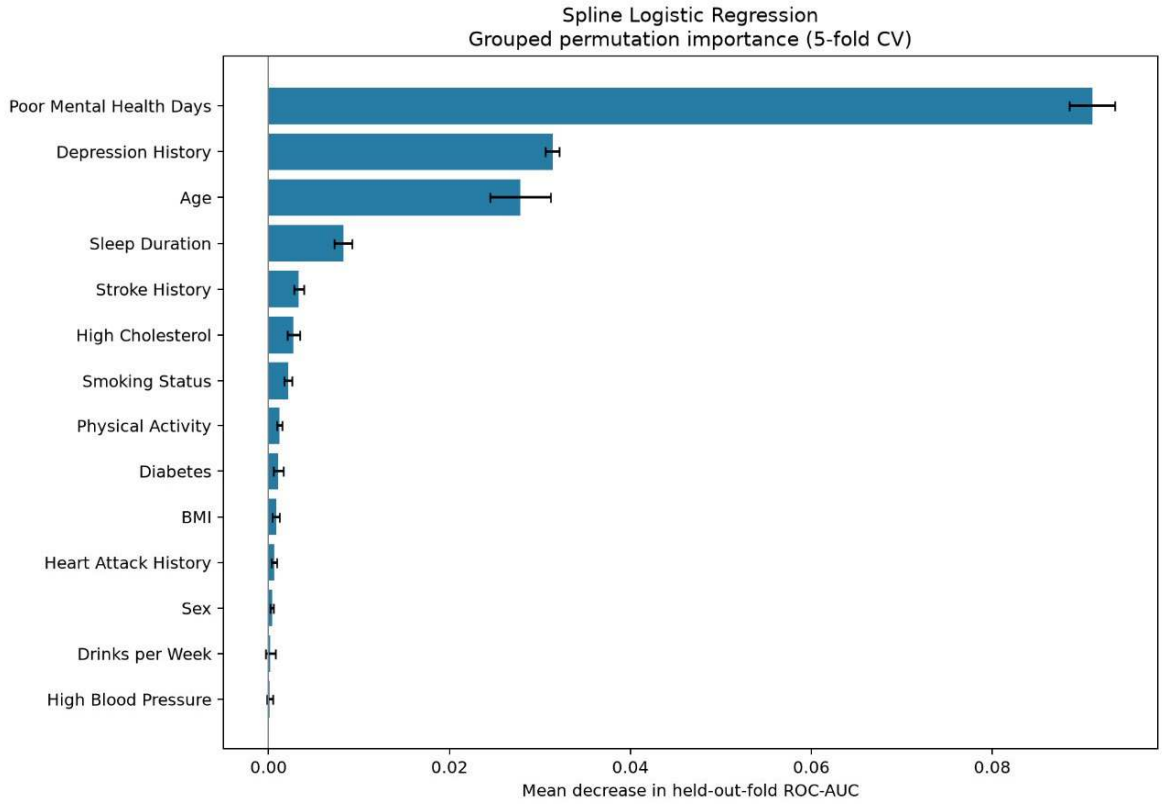


Test set (n = 14,690)	ROC-AUC	AP	Recall	Precision	F1
Spline LR	0.738	0.440	0.705	0.337	0.456
Random Forest	0.733	0.438	0.712	0.333	0.454
Gradient Boosting	**0.738**	**0.446**	0.711	**0.337**	**0.457**
Neural Network	0.733	0.439	0.644	0.358	0.460

Why these metrics

- ROC-AUC: how well the model ranks people; threshold-free (chance = 0.50)
- Average precision : ranking for the rare class; no-skill baseline = 0.21, not 0.50
- Recall: share of people reporting decline we reach (missed people = lost outreach)
- Precision: share of invitations that reach someone reporting decline (outreach efficiency)

Permutation Importance



- Age (0.029) is the strongest non-modifiable factor
- Cross-sectional: Association, not causation
- Mood may worsen because of memory loss

Who to prioritise

Priority	Who?	What does the agency do?
Primary: Screening invitations	Top 4,545 respondents ranked by spline LR predicted scores	Invite them for brief memory screening
Secondary: Tailored support	Screened respondents with relevant health or behavioural needs	Offer lifestyle counselling, appropriate support and follow-up referrals

What characteristics might these respondents have?

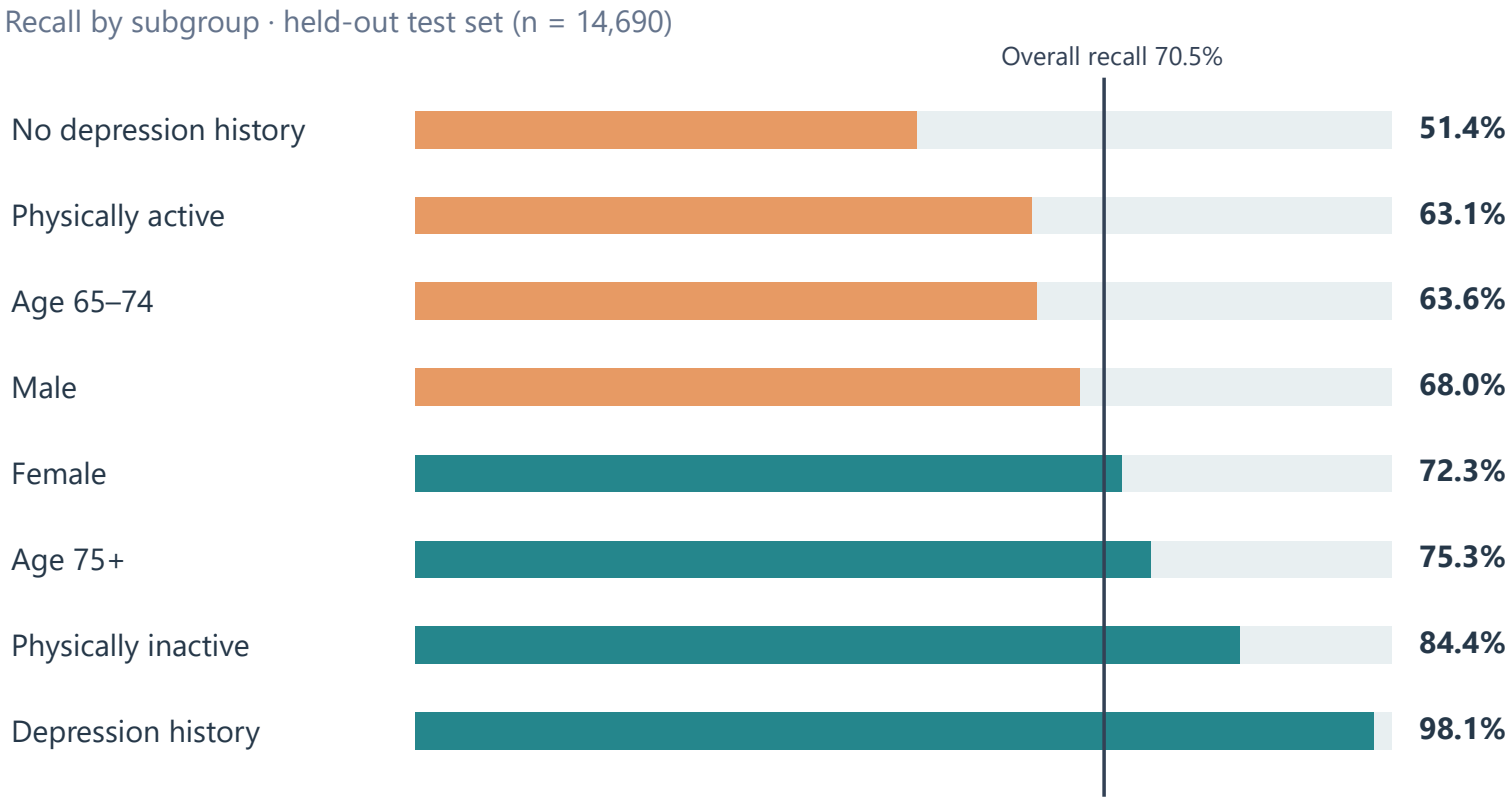
1. Poor mental-health days
2. Depression history
3. Age
4. Sleep duration
5. Stroke history

Which potentially modifiable factors stand out?

Factor	Bucket	What the data shows	Outreach action
Poor mental-health days	Potentially modifiable	8.5 vs 2.8 avg days (decline vs not); top model reliance (0.093 ΔAUC)	Mental-health support, referral
Depression	Manageable clinical	Decline rate 41.0% vs 15.6%; 2nd in reliance (0.031)	Link to existing depression care
Sleep duration	Potentially modifiable	U-shape: short and long sleep both higher; 0.008 ΔAUC	Sleep-health counselling
Physical activity	Potentially modifiable	Inactive respondents report decline more often (9–11 pt gap)	Activity programmes
Smoking	Potentially modifiable	Current smokers higher than never-smokers	Cessation support
Blood pressure, cholesterol, diabetes	Manageable clinical	Small independent contribution once others are included	Care coordination
Alcohol, BMI	Potentially modifiable	Little or no clear association in this data	No targeted action

ΔAUC = drop in ROC-AUC when the variable is shuffled (grouped permutation importance, training folds). Rates are unadjusted EDA figures.

Error analysis | Spline Logistic Regression



Key findings

- Lowest recall: no depression history (51.4%).
- Physically active respondents: 63.1%.
- Age 65–74: 63.6%.

What this means

Lower subgroup recall suggests that some respondents without prominent clinical or behavioural risk indicators may be under-prioritised. Subgroup performance should be monitored before operational deployment.

Outreach opportunity

$5 invitation + $50 brief memory screen = $55 per participant

Capacity: 4,545 participants • Spend: S$249,975

Selection strategy	Positive respondents reached	Precision@4,545	Recall@4,545	Cost / positive
Random (expected)	938.4	20.65%	30.94%	$266.41
Gradient Boosting	1,782	39.21%	58.75%	$140.29
Random Forest	1,776	39.08%	58.56%	$140.77
Spline LR (recommended)	**1,799**	**39.58%**	**59.31%**	**$138.97**
Neural Network	1,767	38.88%	58.26%	$141.48

Recommendation: Pilot spline logistic regression

Why spline LR

- Test ROC-AUC 0.738, approximately equal to Gradient Boosting (0.738)
- 1,799 positive respondents reached vs 1,782 for GB
- Explainable predictor relationships and adjusted curves
- Strong operational yield at fixed capacity

Why not full deployment

- 60% of selected respondents do not report decline
- Subgroup performance needs validation
- Probability calibration not yet established
- Cross-sectional, self-reported data

Pilot conditions

- Illustrative $250k screening budget
- Human oversight of outreach decisions
- Monitor subgroup recall and model drift
- Prospective evaluation before scaling

Risks: Potential subgroup performance disparities; population and survey drift; false positives and missed respondents